11 Information handling, modelling and forward planning

Topic	What students need to learn:
11 Information handling, modelling and forward planning	11.1 - Collection, collation and analysis of information and the use of this to make informed decisions: a) ***marketing*** *– marketing analysis, research techniques, raw data/analysed data to enable enterprise to be encouraged* b) ***innovation management*** *– cooperation between management, designers and production engineers, the encouragement of creativity* c) *the use of* ***feasibility studies*** *on the practicability of proposed solutions.*

a) Marketing – *marketing analysis, research techniques, raw data/analysed data to enable enterprise to be encouraged*

Marketing

- *Marketing Analysis,*
- *Research Techniques*
- *Raw data/analysed data to enable enterprise to be encouraged*

What Is Marketing Explained Definition, Benefits, & Strategies



Marketing Analysis

A ***marketing analysis*** is the part of a business plan or feasibility study that presents information on a specific area of the market in which a company is operating. It is normally in four parts.

1. ***Industry overview*** – a description of the area of the industry that the company is concerned with.

2. ***Target market*** – a detailed view of the demographics of potential customers and their needs. For example, teenagers, disabled people, women in their twenties.

Marketing Analysis

3. ***Competition*** – the current position of competitors in the market and their strengths and weaknesses.

4. ***Pricing and forecast*** – the position that companies want to take up in the market will be dictated by the pricing of the product. The portion of the market that they anticipate will be in the forecast.

Research Design - Choosing your Data Collection Methods

data collection method noun

da·ta col·lec·tion meth·od | \ ˈdā-tə kə-ˈlek-shən ˈme-thəd\

Definition

1 :

Research Techniques

- There are many different ***research techniques*** that can be used by designers.
- Depending on the nature of the data that needs to be collected the type of research to be used will differ.
- In some cases ***raw data*** will be collected and used, whilst in others data will need to be ***analysed*** to encourage enterprise.

We are now going to look at different ***research techniques...***

Research Techniques

Traditional research methods such as *surveys, focus groups and questionnaires* are still used.

The rapid expansion of digital technology and smartphone and tablet ownership has, however, had a significant impact on the way that companies conduct their research.

Research Techniques

New methods facilitated by ***digital technology*** include:

- Targeted surveys and voting carried out using text messaging (SMS)
- Web and application-based surveys specifically designed for smart devices.
- Location-based surveys involving tracking through GPS to obtain localised information.
- Biometric methods which measure responses to stimuli such as TV commercials through methods such as monitoring heart rate, respiration, brain activity.

Research Techniques

- Exploration of virtual environments to provide feedback on things such as potential shop layouts, branding ideas.
- Live video calling to carry out traditional market research techniques more easily.
- QR codes provide a quick way for respondents to answer survey questions.
- Use of social media to collect information such as purchases, trends and interests.

Research Techniques

Observational research methods is becoming more common so that researchers can see what potential customers do rather than what they might say they would do when asked in a survey or focus group.

Data Analysis

Data regarding customer behaviour is now very easily obtained due to the rise in online shopping and the growth of digital data collection methods such as those mentioned previously.

Analysis of data now forms the primary focus and there are many examples of how this takes place:

- Online surveys are usually linked to software that automatically analyses the results and presents the results in a variety of formats, including spreadsheets, graphs and tables.

Data Analysis

- Specialist analysts conduct further detailed analysis to ensure that the most relevant information is extracted.
- Systems such as ‘Google Analytics’ can provide a host of information on website visitors and the pages they have visited.
- Online retailers such as Amazon monitor purchases and provide predictive recommendations.

Data Analysis

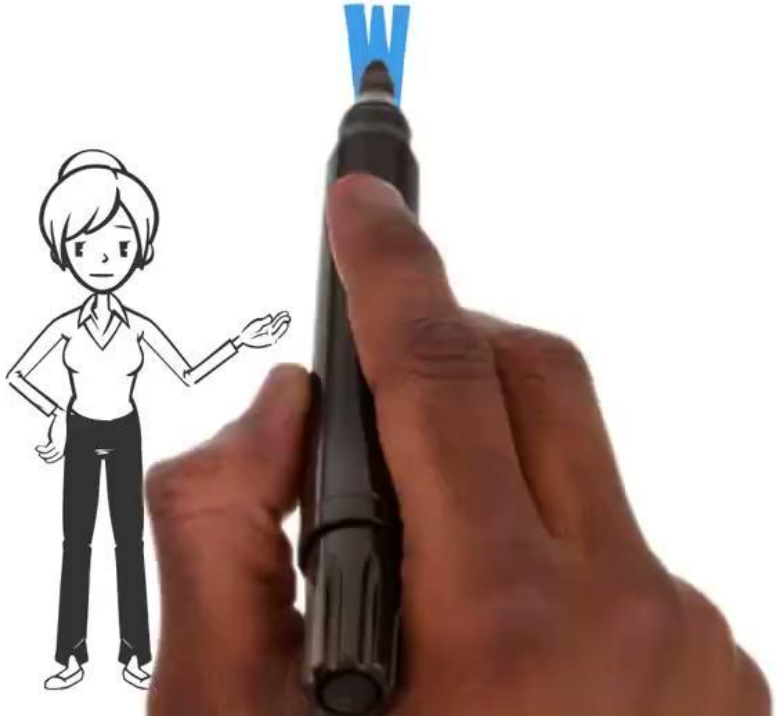
- Data from social media can be harvested, which makes it possible to gauge the feelings, moods, emotions, likes and dislikes of millions of people. This data can be grouped according to age, sex, nationality and many other demographic factors that facilitate detailed analysis.
- The privacy of personal information available on digital platforms is subject to growing concern. As a result of this, government legislation and updated social media company policies are coming into force.

Exam Tip

If you are answering a question on ***market research***, make sure that you cover *how digital technology facilitates access to target data from particular demographic groups.*

b) Innovation Management – ***cooperation between management, designers and production engineers, the encouragement of creativity***

What is Innovation Management?



HYPE

Innovation Management

Innovation management refers to ***the way*** that companies *systematically organise the renewal of the way they work and the outcomes of their activities in order to improve profitability*.

This could, therefore, be the introduction of a new system for Computer-Aided-Design (CAD) or an idea for a completely new product or feature to be added to their range.

Innovation Management

A variety of tools are used by management to realise the creative potential of their design and production team and encourage innovation.

- ***Prototyping*** – this facilitates the rapid exploration of the potential of ideas and has become an increasingly popular method due to the availability of technology such as digital prototyping methods involving CAD and 3D printing.

Innovation Management

- ***Ideation*** – variant of brainstorming, where all members of a design team get to make suggestions in an uninhibited way. The aim is to record everything, with no criticism or negative judgement and there is encouragement to look at situations from a variety of points of view.

Innovation Management

- ***Product Lifecycle Management (PLM)*** – a strategic management technique that focuses on the improvement of time management and reduction of costs through considering all elements of a product from its development through to its manufacture, maintenance, distribution and use. PLM software is used to facilitate profitable innovation, particularly by responding rapidly to current market needs through the adoption of digital design and manufacture.

Innovation Management

- ***Kaizen*** – a Japanese 'incremental innovation' system that involves participation of all tiers of workers in an organisation, from management to shop floor operatives. It encourages the ongoing suggestion, discussion and adoption of improved working practices.
- ***Design Council Support Programmes*** – the Design Council has a range of programmes to support companies in their quest to innovate. A variety of industry specialists work with company teams to augment and fully utilise their skills.

c) the use of feasibility studies on the practicability of proposed solutions

Feasibility Studies of Projects



Feasibility Studies

- The purpose of a ***feasibility study*** is to systematically ascertain the practicality of undertaking the design, manufacture and marketing of a particular product.
- ***Spreadsheets*** play an important role in calculating the two key areas of project costs.
  - ***Direct costs*** – including materials and the labour of manufacturing workers for specific production tasks.
  - ***Indirect costs*** – such as lighting, rent, maintenance and auxiliary labour that is not specific to particular production.

Feasibility Studies

- Sophisticated computer modelling techniques can be used to ascertain the feasibility of particular technical aspects of designs. An example of this is the investigation of a specific process such as injection moulding, bearing in mind factors that dictate potential costs such as:
  - *Initial mould cost and complexity involved in achieving particular features.*
  - *The size of the moulding, type and amount of material required and resulting cycle/cooling time.*
  - *The scale of production required.*

Feasibility Studies

Typical Mistakes: When writing about feasibility studies, don't concentrate on whether a design is a practical proposition for consumers.

Your focus should be on a wider range of issues such as direct and indirect costs, materials, manufacture and marketing.

Stages of a Feasibility Studies

Preliminary analysis of what the study is about

A projected income statement which outlines potential income streams and profitability

A market survey is carried out to establish demand

Planning how the business will need to be organised and the resourcing and staffing of operating

An opening day balance sheet needs to be prepared in order to analyse related costs and the initial total assets and liabilities at the start of the project

All the data that has been gathered is reviewed and analysed to consider the viability and sustainability of the project

A decision is made whether to go ahead or not

Now Test Yourself

1. *Briefly explain **four** ways in which digital systems improve a company's ability to obtain market research data.*
2. *Why is it important for companies to innovate? Explain **three** ways in which a company might improve their management of innovation.*
3. *Why is it important for a feasibility study to be carried out before starting a new project?*
4. *What are the stages of a feasibility study?*
5. *How might modelling or simulation software be used to establish the feasibility of a particular manufacturing method or factory layout?*

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